

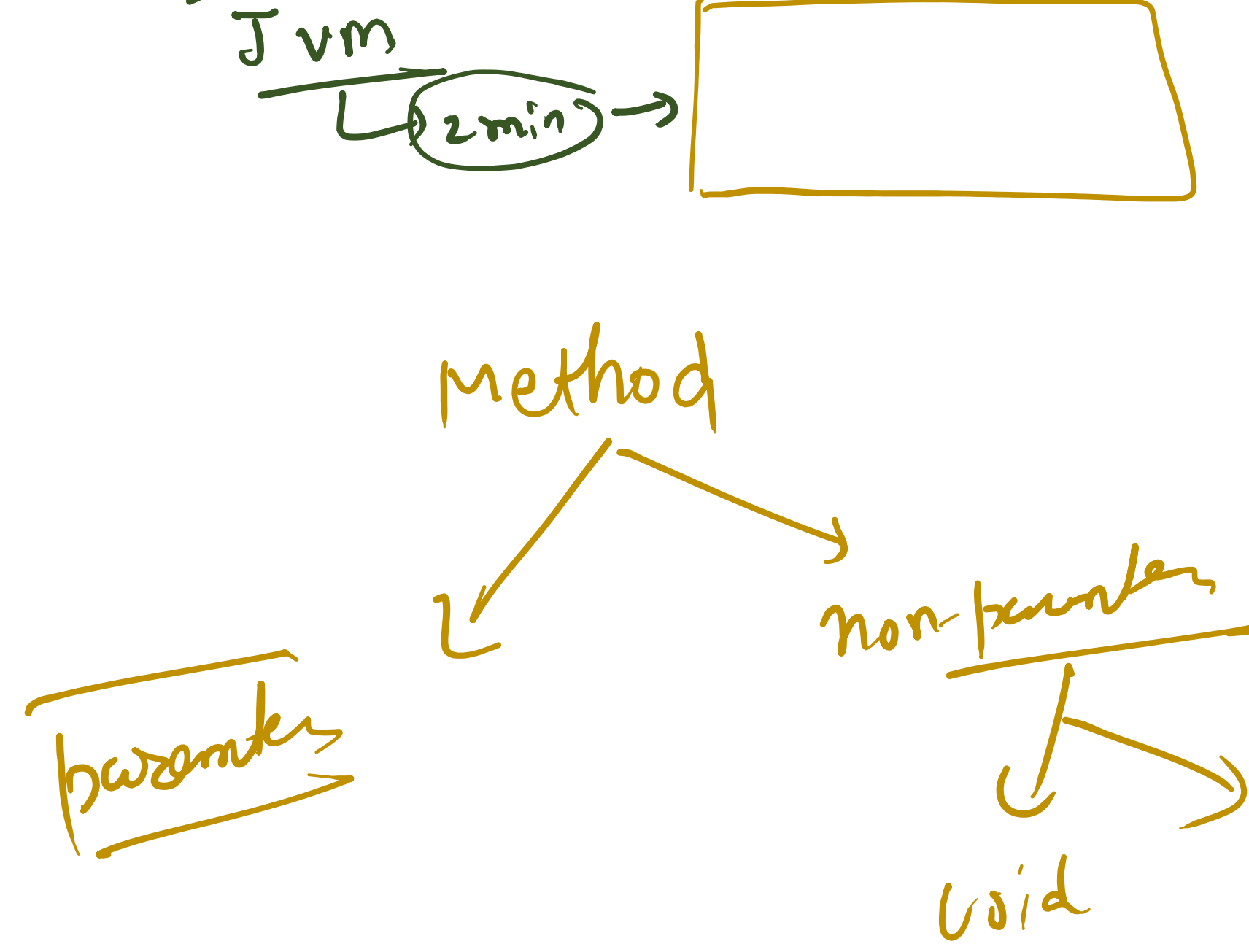
Function/ Method

some $\rightarrow$ input

$f(x) = \text{input} \rightarrow \text{out put}$

$P(n) = x^2 + y$

$\begin{cases} f(-1) = 8 \\ f(2) = 8 \end{cases}$



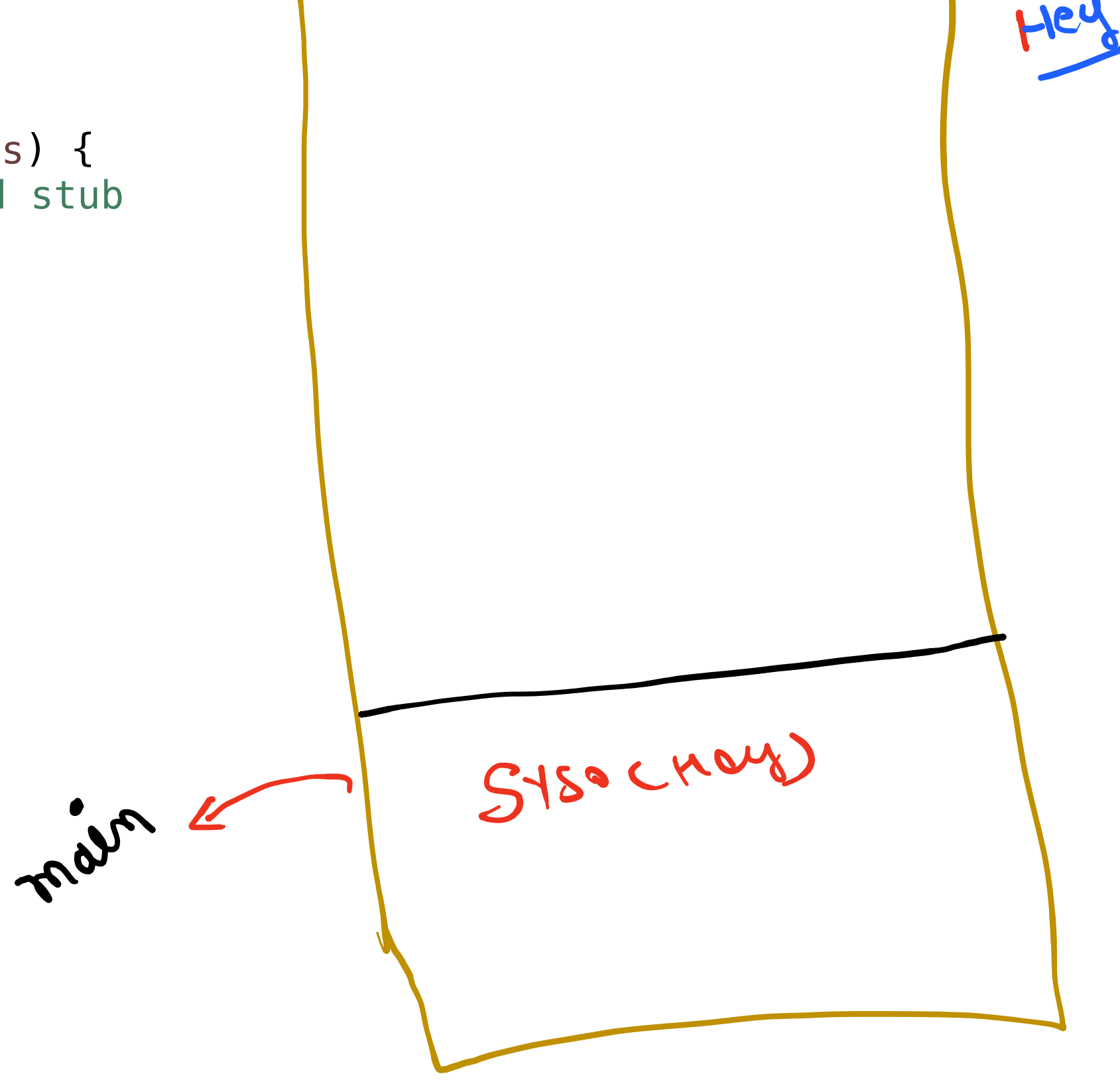
Space

Memory

heap

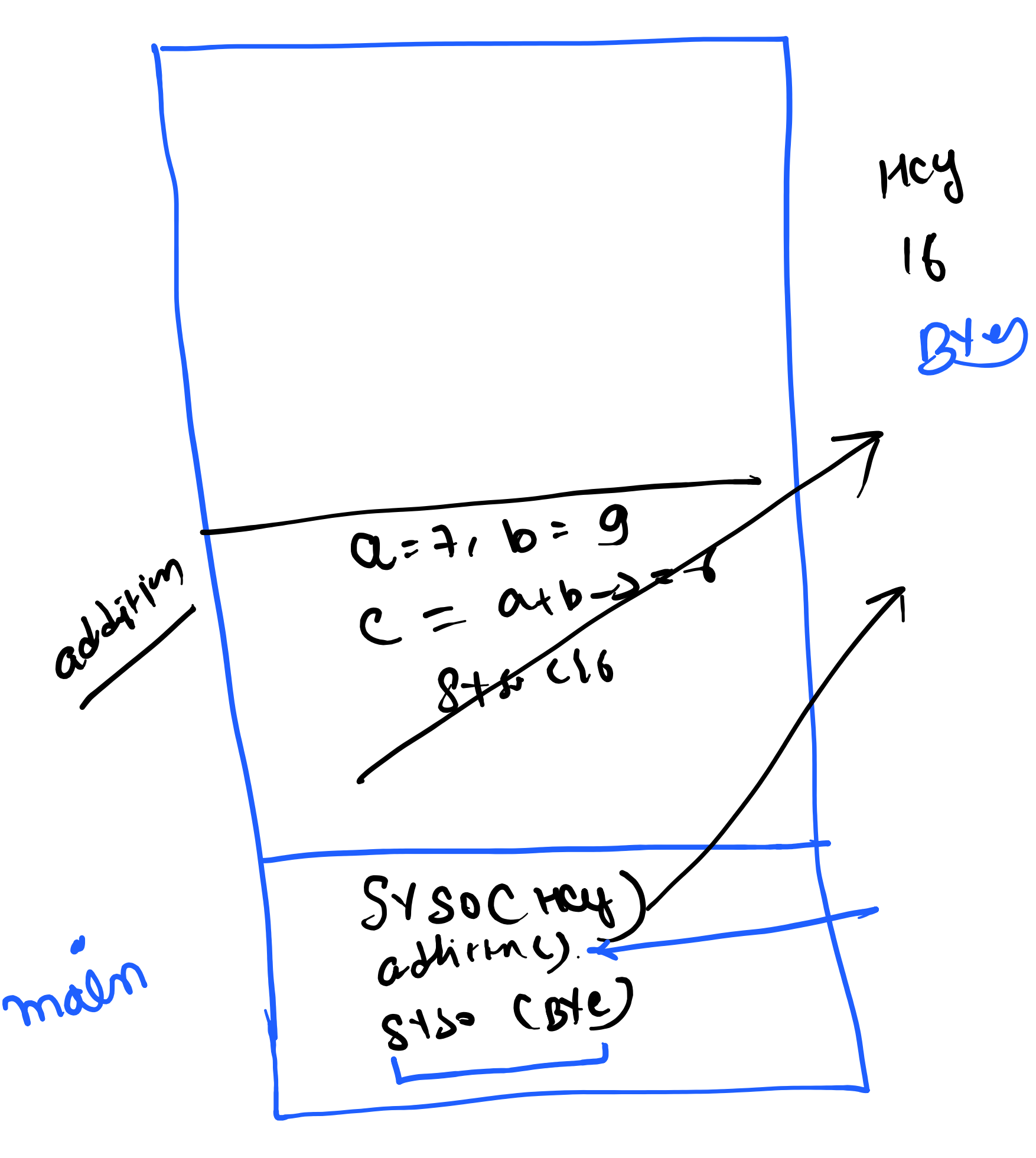
```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.println("hey");
}

public static void Addition() {
    int a = 7;
    int b = 9;
    int c = a + b;
    System.out.println(c);
}
```



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.println("hey");
    Addition();
    System.out.println("Bye");
}

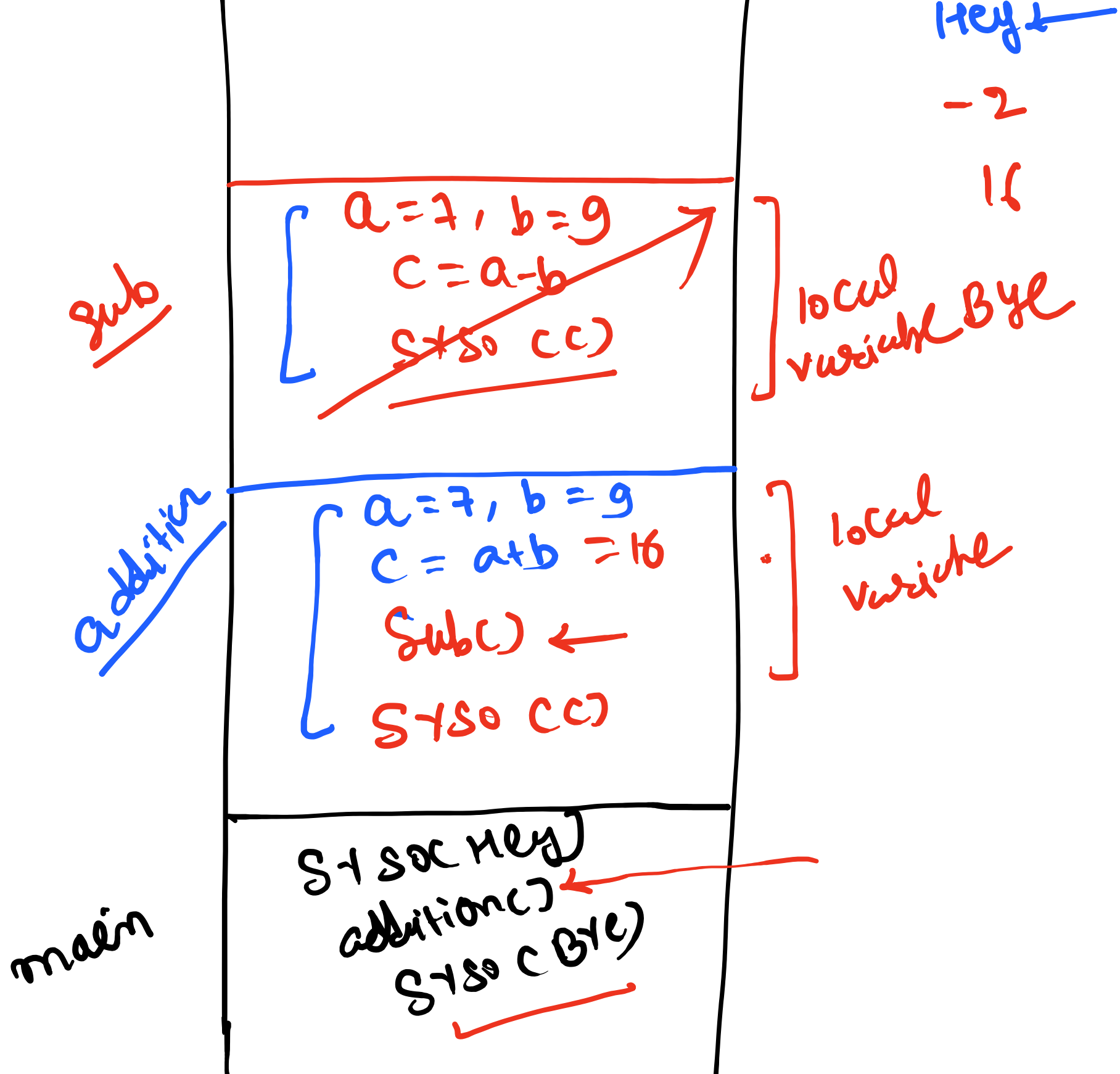
public static void Addition() {
    int a = 7;
    int b = 9;
    int c = a + b;
    System.out.println(c);
}
```



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.println("hey");
    Addition();
    System.out.println("Bye");
}

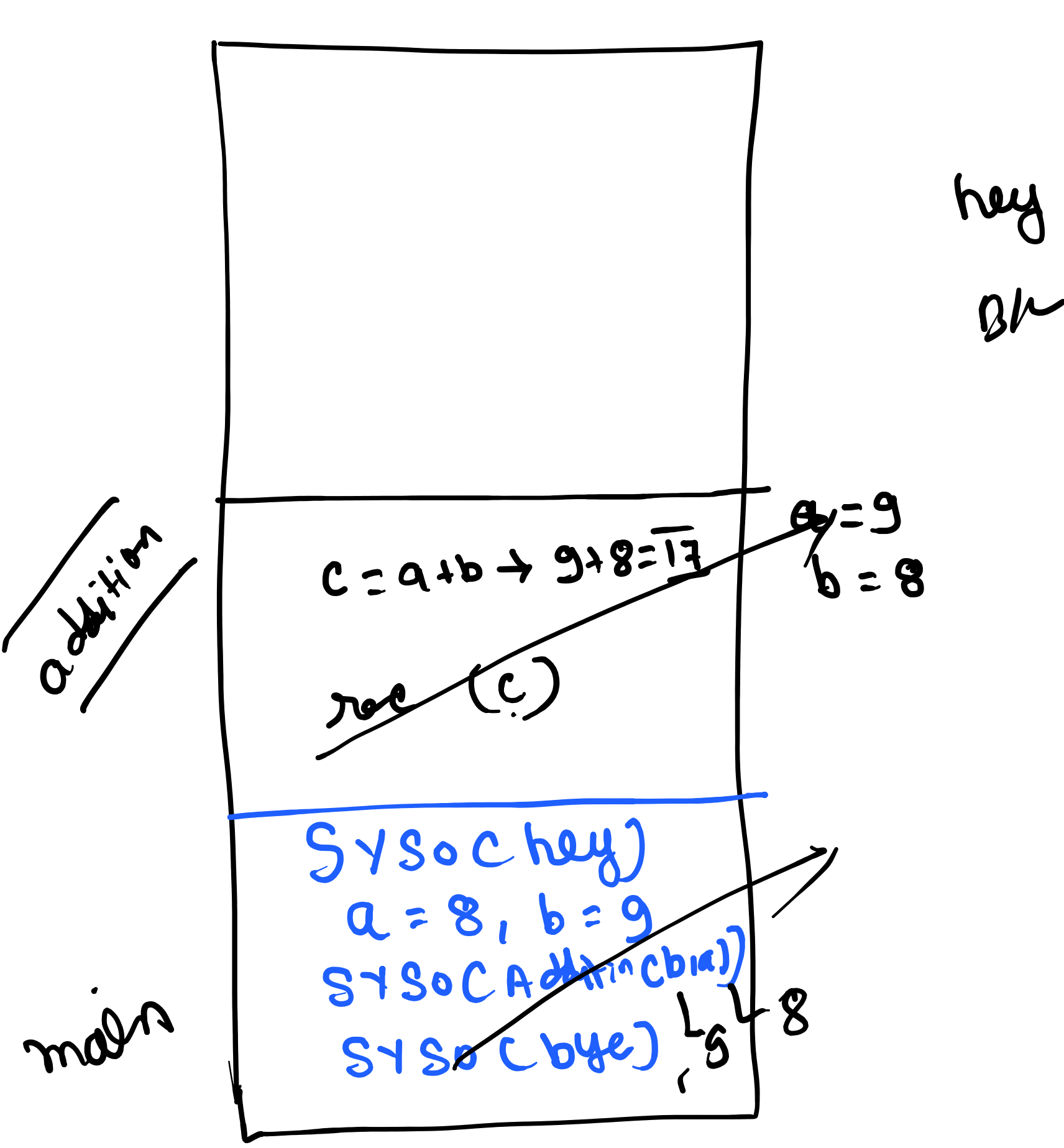
public static void Addition() {
    int a = 7;
    int b = 9;
    int c = a + b;
    Sub();
    System.out.println(c);
}

public static void Sub() {
    int a = 7;
    int b = 9;
    int c = a - b;
    System.out.println(c);
}
```



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.println("hey");
    int a = 8;
    int b = 9;
    // int ans=Addition(b, a);
    // System.out.println(ans);
    System.out.println(Addition(b, a));
    System.out.println("Bye");
}

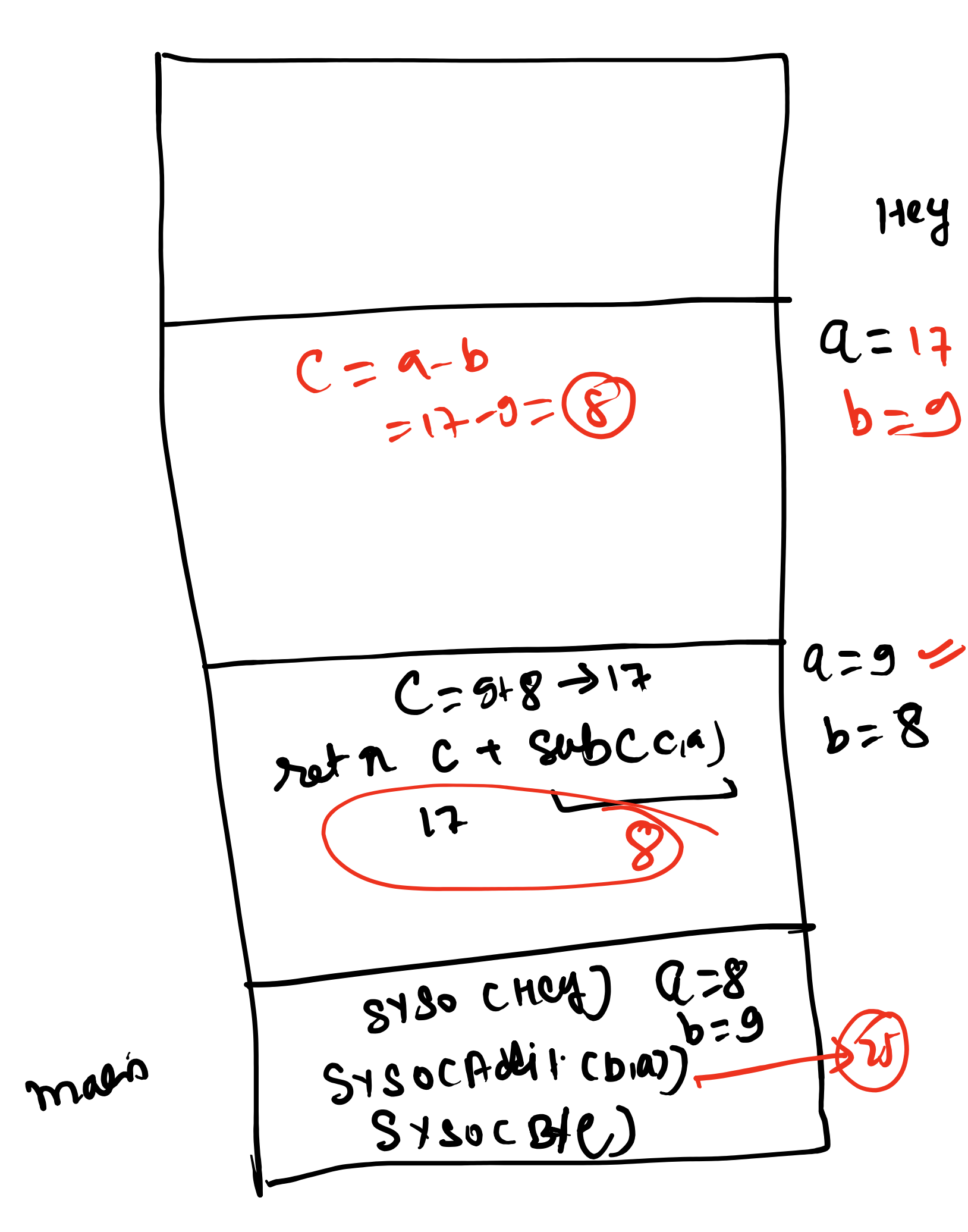
public static int Addition(int a, int b) {
    int c = a + b;
    return c;
}
```



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.println("hey");
    int a = 8;
    int b = 9;
    // int ans=Addition(b, a);
    // System.out.println(ans);
    System.out.println(Addition(b, a));
    System.out.println("Bye");
}

public static int Addition(int a, int b) {
    int c = a + b;
    return c + sub(c, a);
}

public static int sub(int a, int b) {
    int c = a - b;
    return c;
}
```

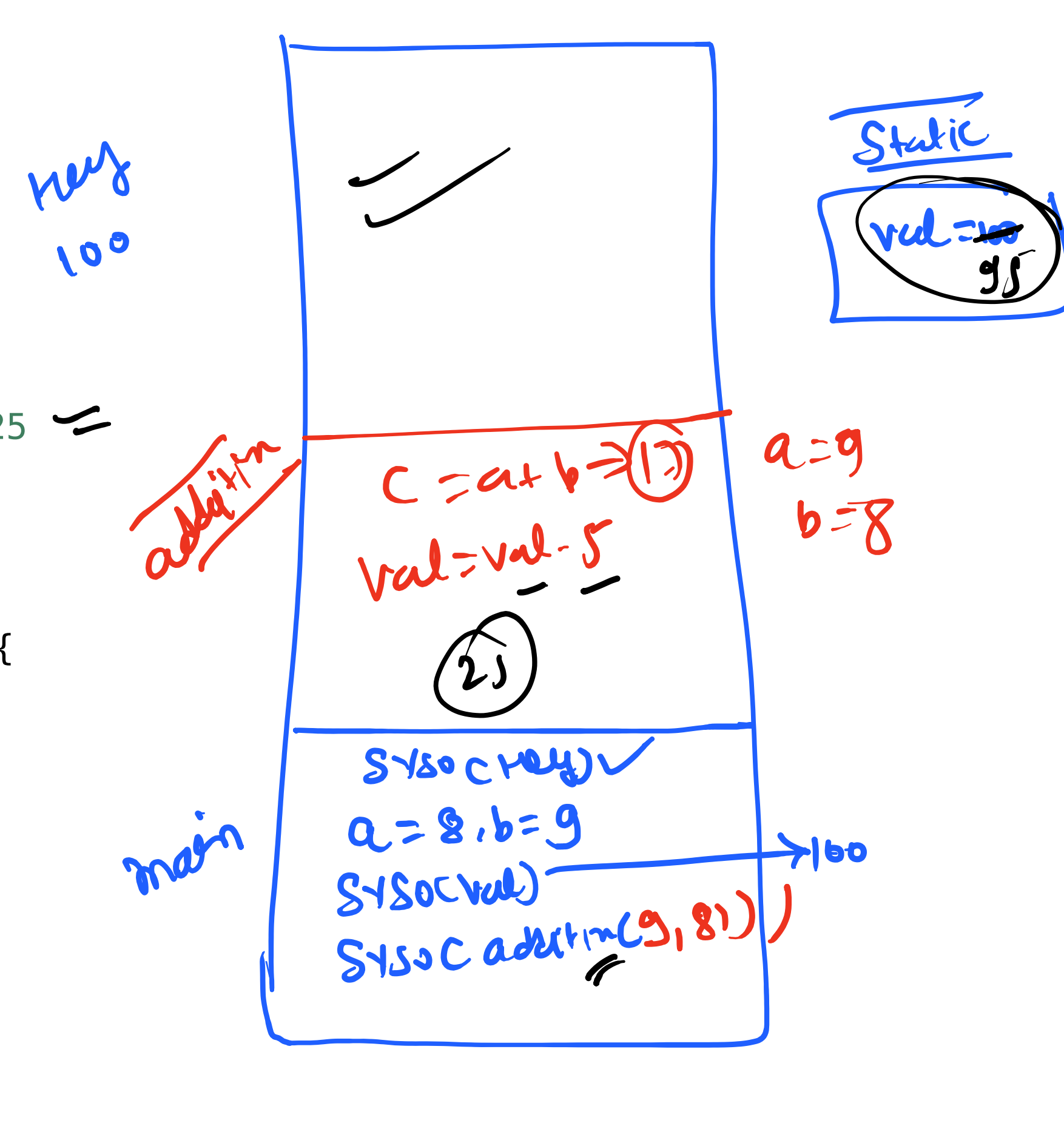


```
static int val = 100;

public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.println("hey");
    int a = 8;
    int b = 9;
    int ans=Addition(b, a);
    System.out.println(ans);
    System.out.println(Addition(b, a));
    System.out.println(val);
    System.out.println("Bye");
}

public static int Addition(int a, int b) {
    int c = a + b;
    val = val - 5;
    return c + sub(c, a);
}

public static int sub(int a, int b) {
    int c = a - b;
    return c;
}
```



```
static int val = 100;

public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.println("hey");
    int a = 8;
    int b = 9;
    // int ans=Addition(b, a);
    // System.out.println(ans);
    System.out.println(val);
    System.out.println(Addition(b, a));
    System.out.println(val);
    System.out.println("Bye");
}

public static int Addition(int a, int b) {
    int c = a + b;
    val = val - 5;
    return c + sub(c, a);
}

public static int sub(int a, int b) {
    int c = a - b;
    return c;
}
```

